
Retail Sales & Profit Analysis

Internship Project Report

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Project Title:

Retail Sales & Profit Analysis Dashboard

Tools Used:

Microsoft Excel

Power BI

Retail Sales & Profit Analysis Report

1. Project Overview

Objective

The objective of this project is to analyze a retail sales dataset, perform data cleaning, extract meaningful business insights, and create an interactive dashboard using Power BI. This project demonstrates key Data Analytics skills including data preparation, business analysis, visualization, and dashboard development.

2. Dataset Description

The dataset contains information related to customer orders, sales performance, product categories, shipping details, customer segments, and profitability.

Key Fields Used:

- Order ID
 - Order Date
 - Sales
 - Profit
 - Discount
 - Shipping Cost
 - Region
 - Product Category
 - Product Name
 - Customer Segment
 - Order Priority
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3. Data Cleaning Process

The following data cleaning activities were performed before analysis:

Duplicate Records

- Checked the dataset for duplicate records.
- No duplicate records were found.

Missing Values

- Missing values were identified in the Product Base Margin column.
- Missing values were replaced using mean imputation.

Date Standardization

- Order Date and Ship Date columns contained mixed date formats.
- Date values were standardized and converted into a consistent date format.

Numeric Data Validation

- Sales, Profit, Discount, Shipping Cost, Unit Price, and Product Base Margin columns were verified and formatted correctly for analysis.

Data Quality Outcome

- Dataset was cleaned and validated successfully.
 - Data became suitable for business analysis and dashboard creation.
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4. KPI Analysis

KPI	Value
Total Sales	14.92M
Total Profit	1.52M
Total Orders	5,496
Average Discount	4.97%
Total Shipping Cost	107.83K

KPI Insights

- The company generated total sales of approximately 14.92 million.
 - Overall profit reached 1.52 million.
 - A total of 5,496 orders were processed.
 - The average discount offered across transactions was 4.97%.
 - Total shipping expenditure was approximately 107.83 thousand.
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5. Business Analysis

Sales by Region

Regional sales analysis revealed significant variation across regions.

Key Findings

- West Region generated the highest sales.
- Ontario and Prairie were the second and third highest-performing regions.
- Nunavut recorded the lowest sales.

Business Insight

The West Region is the strongest market and contributes significantly to overall revenue.

Sales by Product Category

Product category analysis was conducted to identify major revenue contributors.

Results

1. Technology
2. Furniture
3. Office Supplies

Business Insight

Technology products generated the highest sales, indicating strong customer demand for technology-related items.

Top 10 Products by Sales

The highest revenue-generating products were identified.

Leading Products

- Global Troy Executive Leather Low-Back Tilter
- Polycom ViewStation ISDN Videoconferencing Unit
- Canon PC940 Copier
- Riverside Palais Royal Lawyers Bookcase
- Hewlett Packard LaserJet 3310 Copier

Business Insight

A small number of products contribute significantly to overall revenue and should receive strategic attention.

Profit by Customer Segment

Customer segment profitability was analyzed.

Segment Ranking

1. Corporate
2. Home Office
3. Small Business
4. Consumer

Business Insight

The Corporate segment generated the highest profit and represents the most valuable customer group.

Sales by Order Priority

Sales were analyzed across different order priorities.

Observation

- Low and High priority orders contributed the largest share of sales.
- Critical orders generated comparatively lower revenue.

Business Insight

Higher sales are not necessarily associated with critical orders, suggesting that routine business operations drive most revenue.

Monthly Sales Trend

Monthly sales patterns were examined using order dates.

Observation

- Strong sales performance was observed during January, September, October, and December.
- Sales declined during the middle months of the year, particularly around June.
- Revenue increased again toward the end of the year.

Business Insight

Sales exhibit seasonal behavior, with stronger performance during year-end periods.

6. Answers to Project Questions

1. Which region generates the highest sales?

West Region generates the highest sales with approximately 3.6 million in revenue.

2. Which product category is the most profitable?

Technology is the most profitable product category.

3. Which products generate the highest revenue?

The highest revenue-generating products include:

- Global Troy Executive Leather Low-Back Tilter
- Polycom ViewStation ISDN Videoconferencing Unit
- Canon PC940 Copier

4. Which customer segment contributes the most profit?

The Corporate segment contributes the highest profit.

5. What is the monthly sales trend?

Sales show seasonal fluctuations, with stronger performance during January and the final quarter of the year, while mid-year months experience relatively lower sales.

7. Dashboard Overview

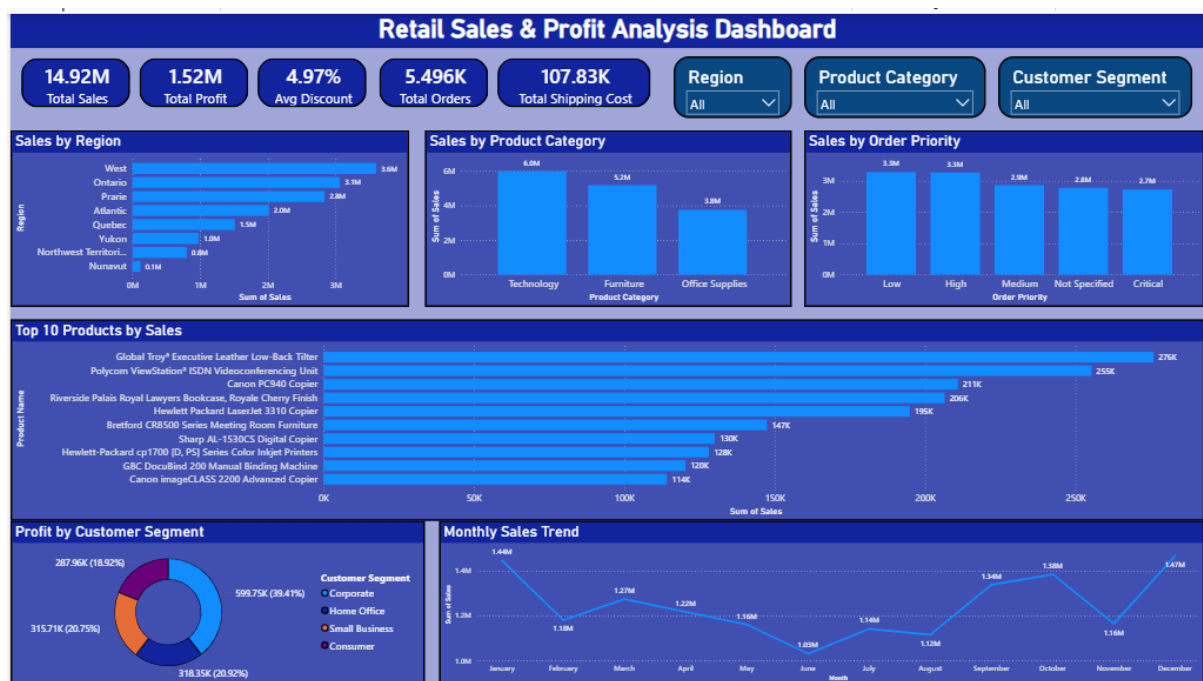


Figure 1: Retail Sales & Profit Analysis Dashboard

An interactive Power BI dashboard was developed containing:

- Total Sales KPI
- Total Profit KPI
- Total Orders KPI
- Average Discount KPI
- Total Shipping Cost KPI

- Sales by Region
- Sales by Product Category
- Top 10 Products by Sales
- Profit by Customer Segment
- Monthly Sales Trend
- Sales by Order Priority
- Interactive Slicers

The dashboard enables users to explore sales performance across regions, product categories, and customer segments.

8. Short Summary of Insights

The retail sales analysis revealed that the West Region is the highest revenue-generating region and Technology is the leading product category in terms of both sales and profitability. Corporate customers contribute the highest share of profits, making them the most valuable customer segment. A small group of products generates a significant portion of total revenue, highlighting opportunities for focused marketing and inventory planning. Monthly sales trends indicate stronger performance during the beginning and end of the year, suggesting seasonal demand patterns. Overall, the business demonstrates strong revenue generation with healthy profitability and clear opportunities for strategic growth.

Conclusion

This project successfully demonstrated the complete Data Analytics workflow, including data cleaning, exploratory analysis, business insight generation, KPI development, and dashboard creation. The findings provide actionable insights that can support business decision-making and improve operational performance.